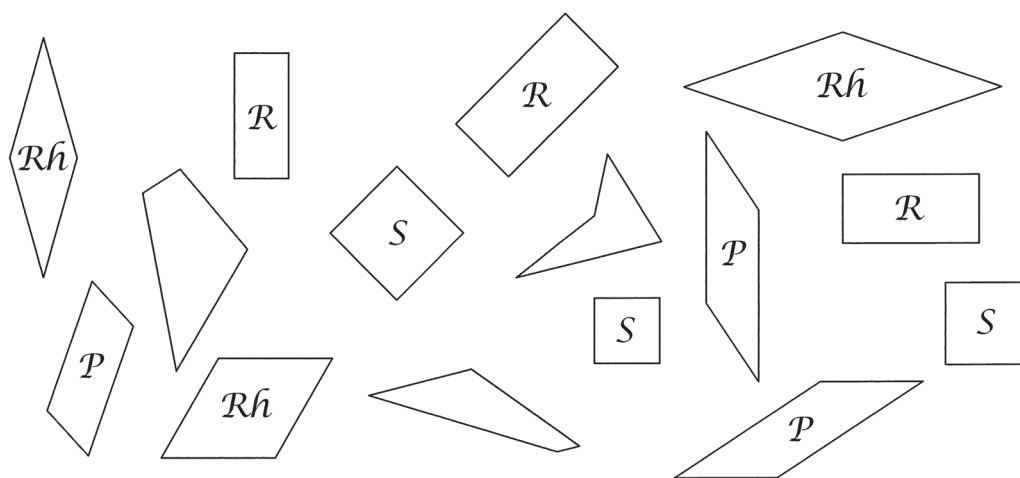


Mathematics is Not Always Cut and Dry

If there's any domain that people think of as having precise and unambiguous concepts, it would have to be mathematics. Here, where contradictions and blurriness should play absolutely no role, one would naturally suppose that the subjective, context-dependent phenomenon of marking, which by definition conflates two categories by assigning them the same label, would surely be nonexistent. And yet this is not the case. Even in mathematics, our human style of fluently jumping between categories, relying on context to make things clear, trumps the desire for pure logicity, as we'll now see.

Tom is in seventh grade. He's just finished a geometry class in which his teacher gave him some homework problems on the topic of quadrilaterals. The first exercise said, "Write 'S' in each square, 'R' in each rectangle, 'Rh' in each rhombus, and 'P' in each parallelogram." Tom diligently carried out his assignment, and the figure below shows what he did.



Tom was very careful not to fall for any trick questions. For example, he wasn't fooled into thinking that the square balanced on one of its corners was a rhombus. And he also correctly identified rectangles that were tilted, and even rhombi that were tilted at strange angles.

The assignment that came next was called "definitions", and it featured the following questions:

1. How do you recognize a square?
2. How do you recognize a rectangle?
3. How do you recognize a rhombus?
4. How do you recognize a parallelogram?

Tom had learned all of this material very well, and he replied as follows:

1. A square has four right angles and four equal sides, parallel in pairs.
2. A rectangle has four right angles and four sides, parallel in pairs.
3. A rhombus has four equal sides, parallel in pairs.
4. A parallelogram has four sides, parallel in pairs.

Tom's teacher graded the homework and when Tom got home, he proudly told his parents that he'd gotten a perfect grade on each of the exercises. Is there anything worrisome in this situation? Well, yes, something is wrong; in fact, there are interesting contradictions here. Indeed, what would have happened if Tom's acts of writing letters inside shapes had been guided by the written answers that he gave?

A square has four right angles and four equal sides, parallel in pairs. So far so good. This works for all the squares in which Tom wrote 'S', and no other figure is described by the phrase.

A rectangle has four right angles and four sides, parallel in pairs. Here things are a little trickier. Tom looked for figures having four right angles and sides parallel in pairs. And yes, the rectangles in which he wrote 'R' all satisfy this criterion, even the tilted ones that almost fooled him — but the squares with 'S' in them are *also* described by this phrase. So why didn't he write both 'R' and 'S' in all the squares?

A rhombus has four equal sides, parallel in pairs. This category also is tricky. That is, all of Tom's rhombi with 'Rh' in them are indeed described by this definition, but the squares, once again, are also described by it. So why didn't Tom write 'Rh' in each square (as well as 'R' and 'S')?

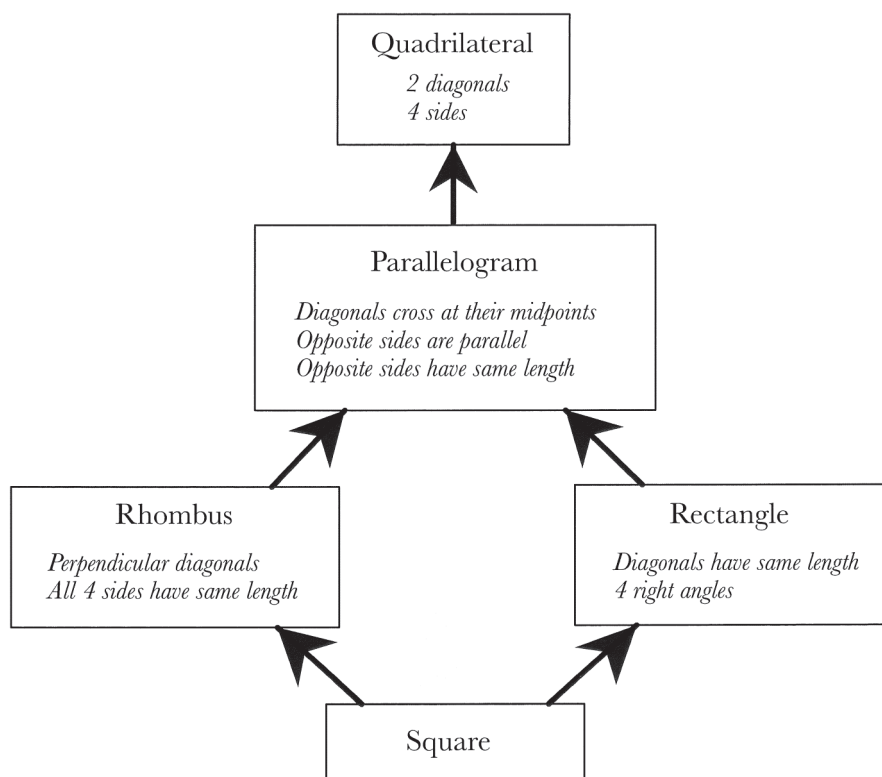
A parallelogram has four sides, parallel in pairs. The plot thickens... Of course, every figure in which Tom wrote 'P' satisfies this criterion, but nearly all of the other figures do too: all the rhombi satisfy it, as do all the rectangles and squares. And so, if Tom's placement of letters had been consistent with his written answers, he would have had to put a 'P' in twelve of the figures, indicating that all but three were parallelograms.

Why, then, did Tom's teacher give him a perfect grade, when in fact his answers to her two assignments have just been shown to be inconsistent?

Are Squares Rectangles?

Indeed: are squares rectangles? The answer comes down to a classic case of marking. From a mathematician's point of view, a square is certainly a rectangle, because it satisfies the criteria that define rectangles. In that sense, the question is unambiguous, and the answer is simply "yes".

If one looks at the definitions of the various types of quadrilaterals, one can easily draw a diagram showing their relationships:



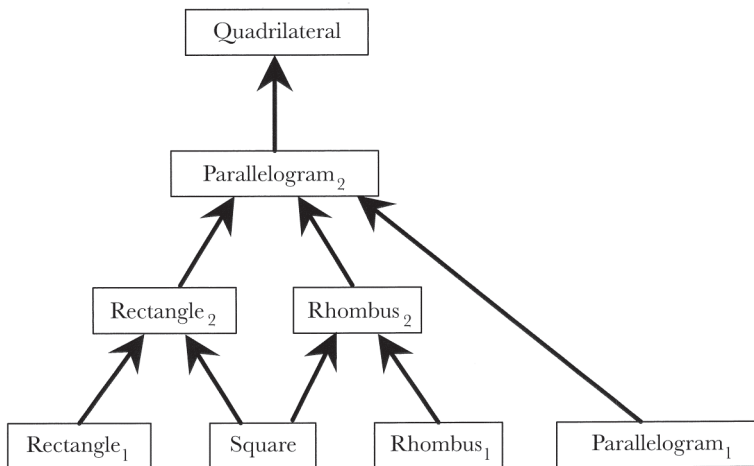
The different categories shown in the figure illustrate the different points of view that one can adopt for any of the quadrilaterals drawn. According to the diagram above, a square belongs to five different categories: *square*, *rectangle*, *rhombus*, *parallelogram*, and *quadrilateral*. If it's seen as a quadrilateral, a square has two diagonals, but there's nothing special to say about them. If it's seen as a parallelogram, a square's two diagonals slice through each other exactly in their midpoints. If it's seen as a rhombus, its diagonals are perpendicular; and if it's seen as a rectangle, they have the same length as each other.

And yet this chart of categories in the world of quadrilaterals isn't sophisticated enough for us to anticipate how Tom's geometry teacher would react if one of her pupils were to draw nothing but a single square as the answer to the following exercise: "Draw a square, a rectangle, a rhombus, and a parallelogram." Strictly speaking, she should be very respectful of the precise understanding of the definitions implicit in such an answer, and she should give the young showoff the highest possible grade. So let's suppose that she was a good sport, and that she did so. Even so, she would have to be taken by surprise by this playful answer, and depending on her personality, she might be either charmed or exasperated by the highly unusual interpretation of her question. But why would she be so surprised to see such an answer, given that it's perfectly correct and elegantly economical, to boot?

Our chart of quadrilateral types, given above, doesn't allow us to predict how Tom's teacher would react if Tom, on his first assignment, were to write four different labels in each square (indicating that it is simultaneously a square, a rectangle, a rhombus, and a parallelogram), and if he were to write two labels in each non-square rectangle (indicating that it is both a rectangle and a parallelogram), and if he were also to write two labels in each non-square rhombus (indicating it is both a rhombus and a parallelogram). Such labels would come directly from a careful application by Tom of his own definitions (in his second answer) — definitions that were so warmly applauded by his teacher and that were given top grades. Would she have given him full credit for answering exercise #1 in such an unorthodox fashion, when we know that in fact she gave him full credit for a totally different, much more conventional set of answers? One would hope that she would be delighted to find in her class a student who is so insightful, and that such thoughtful answers would be rewarded with full credit, and without reservation — but that would depend on whether she appreciated or disdained unconventional viewpoints.

Indeed, categories similar to *non-square rectangle* — for example, *non-square rhombus* and *parallelogram that is neither a rectangle nor a diamond* (and thus not a square either) — are needed to bridge the gap between rigorous mathematical definitions and informal human concepts. To be more precise, we tend unconsciously to presume “non-square” when we think “rhombus” — and yet square rhombi exist, of course. However, they are such a special case that we have to distinguish them from “normal” rhombi, which are not square. Once again, we find ourselves square in the territory of marking. For example, we could posit two new categories, *rhombus*₁ and *rhombus*₂, analogous to *car*₁ and *car*₂. Just as the marked category *car*₁ excludes trucks while the unmarked category *car*₂ includes them, *rhombus*₁ would exclude squares while *rhombus*₂ would include them.

Every square is thus a member of *rectangle*₂ but not necessarily of *rectangle*₁, and every rectangle (whether of type 1 or 2) is a member of *parallelogram*₂ but not necessarily of *parallelogram*₁. And so we see that the various types of quadrilaterals fall into a more complex diagram than the one we saw earlier. It looks as follows:



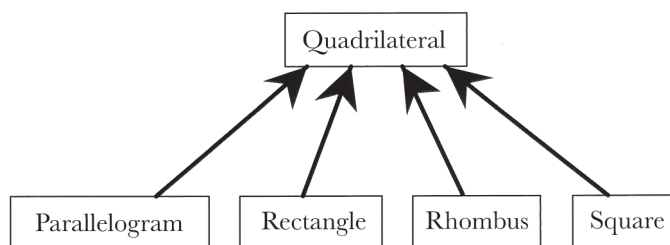
There are now eight categories, organized on four levels of abstraction; all eight are needed to capture what we might call “expert knowledge”, which comes from two sources: first, an understanding of rigorous and formal mathematical definitions, and second, a sense for how these words are actually used in various contexts by someone who has fully mastered this small but slippery mini-domain of math. It’s not hard to see why this can give schoolchildren headaches.

We aren’t saying that this kind of hierarchical diagram is an explicit structure (like the sequence of letters of the alphabet) that gets consciously committed to memory by all people who understand quadrilaterals well. To the contrary, no memorization is involved at all. If one has a mathematical bent, each link in the diagram comes directly from a clear understanding of how two particular concepts are related, and the consequences flow out almost trivially. And yet ironically, this kind of crystal-clear understanding rests tacitly on an ambiguous way of using words: sometimes a square is seen *in contrast* to a rectangle, while other times it is seen as *a kind of* rectangle.

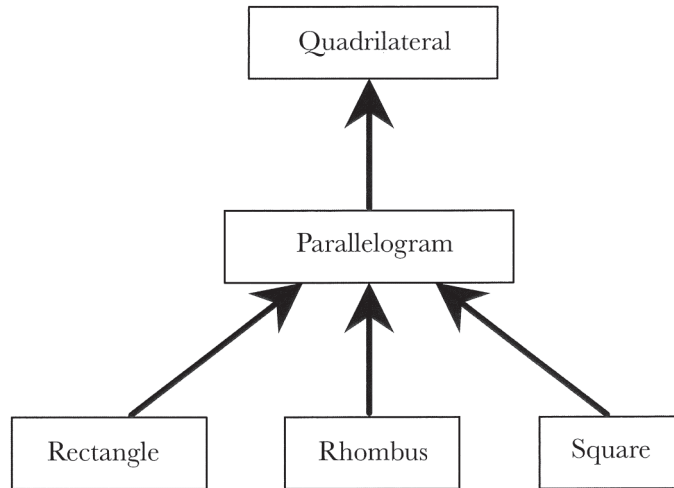
All this goes to show that even in mathematics, where utter precision is expected, it’s commonplace to have terms that stand for two or more concepts, hence are ambiguous. In this particular case, to be an expert in the domain, one needs to have various categories such as *non-square rectangles* (that is to say, *rectangle*₁), and *rectangles whether square or not* (that is, *rectangle*₂). Altogether, the number of such terms comes to eight, which exceeds the number of lexical items by three. And it is by no means easy for people to acquire the proper organization of the categories associated with the words “square”, “rectangle”, “rhombus”, “parallelogram”, and “quadrilateral”.

Indeed, a study we undertook in France showed that the majority of university and middle-school students are unaware of how some of these categories include others, and many simply refuse to accept the idea that squares are rectangles or rhombi. They may go so far as to invent new properties, if one asks them for definitions. For example, they may insist, “A rectangle has to be wider than it is high” or “A rectangle has two pairs of equal-length sides, but not all four sides have the same length”, or then again, “A rhombus has four equal-length sides, but it has no right angles”.

This reveals that the most general interpretation of these words — that is, the unmarked sense — is usually not recognized, and that students’ improvised definitions mostly describe *marked* categories (such as *non-square rectangle*, *non-square rhombus*, etc.). The most frequently observed idea in the minds of these students, all of whom had supposedly mastered all of these notions (indeed, these matters were very fresh in the minds of the middle-school participants, and were assumed to be part of the background knowledge of the undergraduate participants), is just a two-tier hierarchy with *quadrilateral* on top, and *parallelogram*, *rectangle*, *rhombus*, and *square* below.



Students who gave slightly more sophisticated answers tended to add just one middle level — *parallelogram* — which they interpolated between the top level (where *quadrilateral* was found) and the bottom (on which resided *rectangle*, *rhombus*, and *square*):



Our investigations showed that an expert-like hierarchy having *four* levels (as in the figure two pages back) was present in only a very small percentage of the participants.

As we have just seen, part and parcel of becoming an expert in a domain involves, over and above possessing many categories, organizing them efficiently. Thus, to understand quadrilaterals well, one has to know a good deal more than just the fact that squares, rectangles, rhombi, and parallelograms are all special types of quadrilaterals; one also has to know the relationships that interconnect them. This theme — that an efficient organization of categories is part and parcel of expertise — is important, because expertise is far from being limited to narrow, technical domains. Expertise is what anyone acquires who has deep knowledge of *any* domain; everyone is an expert in their everyday environment, as well as in their profession and in their various hobbies. To be sure, expertise does not always require high degrees of creativity or insight, although of course it doesn’t exclude those.

The Verticality of Expertise

You might aspire to become a past master in dog breeding, or perhaps in the etymology of English words, or in the varieties of tea, or in the world of banking, or in Spanish literature, or opera or history or anatomy... If you succeed, some will call you a “walking encyclopedia” of your specialty. Mr. Martin takes great pleasure in spending afternoons reading about dogs. Aside from learning the names of many breeds, including “poodle”, “bulldog”, “German shepherd”, “basset”, “chihuahua”, “golden retriever”, “grayhound”, “pit bull”, “borzoi”, “Saint Bernard”, “fox terrier”, “Dalmatian”, “doberman”, “labrador”, “dachshund”, “spaniel”, and so on, he has